Bhavesh_Chaudhari

Synopsis: I am a Python & PySpark Data Engineer with more than 3 years of experience in building ETL pipelines, processing large-scale data using the Hadoop ecosystem (Spark, Hive, HDFS), orchestrating workflows with Apache Airflow, working with databases, and using AWS. I am skilled in improving data quality and reliability and creating analytical solutions and Generative AI on AWS.

EXPERIENCE SUMMARY

I have over 3.5 years of experience in AWS cloud data engineering, utilizing Python, PySpark, and Hadoop-based frameworks to build and manage scalable data pipelines.

- ➢ Strong expertise in Python and PySpark for large-scale distributed data processing across Hadoop and cloud-native environments.
- ➢ Proficient in PySpark and SQL for complex data transformations and query optimization on distributed datasets, with strong fundamentals in SQL for analytical and relational workloads.
- ➢ Orchestrated complex, multi-step data workflows using Apache Airflow (AWS MWAA), enabling reliable pipeline scheduling and dependency management.
- ➢ Developed analytics solutions to deliver actionable insights and deployed Gen AI solutions on AWS.
- ➢ Designed and implemented scalable, cost-effective data architecture on AWS.
- ➢ Hands-on experience with AWS services including AWS Glue, S3, Lambda, Redshift, Athena, Step Functions, DynamoDB, RDS, IAM, CloudWatch and SNS.
- ➢ Designed and implemented scalable and cost-effective data architecture on AWS, enhancing data quality and reliability.

Cloud Computing Skills

Data Languages

→ Python
→ SQL
→ PySpark

Big Data Technology

→ Apache Hadoop (HDFS)

Development Tools

→ Apache Airflow / MWAA
→ VS Code
→ GIT
→ Jupyter Notebook

AWS Services

→ AWS Lambda
→ AWS Glue
→ Amazon Redshift
→ AWS DynamoDB
→ AWS SNS
→ QuickSight

RDBMS

→ MYSQL

GCP Services

→ BigQuery
→ Dataproc (Hadoop/Spark)
→ Google Function run
→ Pub/Sub
→ Dataflow
→ Firestore

QUALIFICATION

Degree	Institute	Percentage	Year
Bachelor of Engineering (B.E)	AISSMS IOIT	83.03	2022
Intermediate	Bytco College	81.85	2018
SSC	P.E.School	91.20	2016

CERTIFICATION

Certification	Institute	Year
Cloudera on Hadoop Certification	Cloudera	2026
AWS Certified Cloud Practitioner	AWS	2023
Cloud Digital Leader	GCP	2024

Working with Hexaware Technologies Ltd. From May 2022 – Till date

PROJECT EXPERIENCE

Project Name : D&G

Role : ETL Developer and Data Engineer

Environment : AWS

Duration : SEP 2023 to SEP 2023

Tools Used : AWS Glue, Glue Data Quality, S3, Lambda, CloudWatch Events, AWS QuickSight, Redshift, Python

Description:

I worked with the project team to create a data validation and audit framework using AWS services. Python scripts were developed to orchestrate data quality checks and automate validation logic. We integrated AWS Glue for Glue Data Quality to maintain data integrity and accuracy in the data pipelines. AWS Lambda and CloudWatch Events were used for automated data validation, improving efficiency and reliability. AWS QuickSight was implemented to build a detailed Data Quality Dashboard, offering real-time insights into data quality metrics for stakeholders. The project aimed to streamline data validation, ensure data accuracy, and provide insights for informed decision-making. I collaborated closely with stakeholders to meet project needs and align with business requirements.

Roles and Responsibilities:

- Collaborated with the project team to develop a data validation and audit framework using AWS services and Python.
- Developed Python-based validation scripts to automate data quality checks and improve pipeline reliability.
- Integrated AWS Glue for Glue Data Quality to ensure data integrity and accuracy.
- Utilized AWS Lambda and CloudWatch Events for automated data validation processes, enhancing efficiency and reliability.
- Implemented AWS QuickSight to develop a comprehensive Data Quality Dashboard, providing insights for stakeholders.
- Worked closely with stakeholders to address specific project needs and ensure alignment with business requirements.

Project Name : Marriott

Role : Data Engineer

Environment : AWS

Duration : OCT 2023 to OCT 2023

Tools Used : AWS Lambda, API Gateway, S3, OpenAI GPT, LangChain, Amazon Bedrock, HTML, CSS, JavaScript, Python

Description:

I developed and demonstrated a Generative AI solution for the hospitality industry, specifically for Marriott, by integrating AWS Lambda with OpenAI's GPT language model using LangChain and AWS Bedrock for advanced AI capabilities. Python was used as the core programming language for all backend logic, prompt engineering, and LLM integration. I used API Gateway for secure API management and S3 for scalable data storage. AWS Lambda handled cost-effective, serverless processing and integration with the AI models. The project demonstrated how AI can improve customer experiences and operational efficiencies in the hospitality industry. It involved developing a complete frontend using HTML, CSS, and JavaScript for a seamless user experience. The AI solutions were tailored to meet industry-specific requirements and align with business goals, offering robust and scalable enhancements to hospitality services.

Roles and Responsibilities:

- Developed AI solutions in Python, integrating AWS Lambda with OpenAI GPT LLM using LangChain and AWS Bedrock for advanced AI capabilities.
- Utilized API Gateway for secure API management and S3 for scalable data storage.
- Employed AWS Lambda for cost-effective, serverless processing and integration with AI models.
- Developed the complete frontend using HTML, CSS, and JavaScript for a seamless user experience.
- Showcased how AI can enhance customer experiences and operational efficiencies.
- Ensured the solutions met industry-specific requirements and business goals.

Project Name : Delta Airlines (Informatica to AWS Migration)

Role : ETL Developer and Data Engineer

Environment : AWS, INFORMATICA

Duration : JULY 2024 to DEC 2024

Tools Used : AWS Glue (PySpark), Lambda, DynamoDB, S3, SNS, Apache Airflow (MWAA), Secret Manager, Visual Studio Code, ServiceNow, GitHub

Description:

The Delta Airlines project focused on moving their old data processes, which were built using the Informatica platform, to more modern AWS tools. The goal was to make their data systems faster, more scalable, and cost-effective. To do this, the team carefully analyzed the existing workflows and migrated them to AWS services. PySpark on AWS Glue was used for large-scale data transformation and processing. Apache Airflow via AWS MWAA was used to orchestrate end-to-end workflows, manage task dependencies, and schedule pipeline runs. Lambda was used for running lightweight serverless operations, S3 for storing data, and CloudWatch to monitor everything. AWS CDK was used to manage and set up resources automatically, making the system reliable and easy to expand. The data came from various places, such as databases, flat files, and COBOL-based mainframe systems, and was stored in a Teradata data warehouse. This migration enabled Delta Airlines to modernize its data architecture, improve data accessibility and support better decision-making across its operations.

Roles and Responsibilities:

- ➢ Reviewed and analyzed the existing data workflows built with Informatica and migrated them to AWS tools using Python and PySpark to improve performance, scalability, and efficiency.
- ➢ Built and designed PySpark-based ETL pipelines on AWS Glue to move and process data from source databases into Teradata.
- ➢ Orchestrated data pipelines end-to-end using Apache Airflow (AWS MWAA), managing task scheduling, dependencies, and failure alerting.
- ➢ Used Git to manage code changes and ensure an organized development process.
- ➢ Performed thorough testing to check data accuracy and ensure workflows worked properly, creating detailed test cases to maintain data reliability.
- ➢ Deployed workflows to the production environment with minimal disruption and supported operations after deployment to resolve any issues.
- ➢ Documented workflows, ETL processes, and testing steps to make future updates and maintenance easier.

Project Name : Carlyle

Role : ETL Developer and Data Engineer

Environment : AWS

Duration : JAN 2025 to JULY 2025

Tools Used : AWS Glue (PySpark), Glue Data Quality, S3, Lambda, CloudWatch Events, AWS DynamoDB, VS Code, AWS Athena, AWS StepFunction

Description:

I am currently working on a project called the Metadata Ingestion Framework, which involves three main tasks: data extraction, data ingestion, and data integration. Python and PySpark are the core technologies used for all transformation and data processing logic. In the data extraction phase, we copy data from various sources to our centralized hub on AWS S3. During data ingestion, we read the data stored in the S3 bucket using PySpark, ensure it is not corrupted, and convert it to Parquet format in another S3 bucket called the bronze layer. Finally, in the data integration phase, we perform data quality checks and, based on customer requirements, move partial or full data to the silver layer S3 bucket. Apache Airflow (MWAA) is used to schedule and orchestrate the entire pipeline. We use DynamoDB to store metadata such as bucket information, source attributes, paths, load details, and manifest file details etc., which is why it is called the Metadata Ingestion Framework.

Roles and Responsibilities:

- ➢ Create and implement the Python and PySpark-based logic for the ETL (Extract, Transform, Load) pipeline.
- ➢ Orchestrate the end-to-end data pipeline using AWS StepFunction for scheduling and dependency management.
- ➢ Use AWS CloudWatch to monitor the pipeline and resolve any errors.
- ➢ Store all job log information and error details in Iceberg tables within the data catalog.
- ➢ Store job and error log details in a bookmark table in DynamoDB.
- ➢ Send notifications of errors through SNS (Simple Notification Service) topics.

PoC Work Experience

1. **Gen AI Solution for Legal Case Citation and Summary Generation**

- ➢ Developed a Gen AI solution on AWS using Python, focusing on legal case citation and summary generation.
- ➢ Leveraged AWS Bedrock to create a PoC demonstrating tailored solutions for complex legal challenges.
- ➢ Utilized API Gateway for secure API management and S3 for scalable data storage.

- ➢ Employed AWS Lambda for cost-effective, serverless processing and integration.
- ➢ Showcased the potential of AI to streamline legal processes and enhance operational efficiency.

2. **Assisted in developing a PoC focused on AWS DataZone.**

- ➢ AWS DataZone facilitates data management, governance, and access control, enhancing data collaboration across the organization.
- ➢ Implemented features for cataloging, searching, and sharing data assets securely within the organization.
- ➢ Utilized DataZone's capabilities to streamline data discovery, ensuring compliance with data governance policies.
- ➢ Demonstrated how AWS DataZone can optimize data workflows and improve decision-making processes in response to market needs.

3. **PoC Development for Statement of Work (SOW) Generator using AWS Bedrock**

- ➢ Developed a PoC using Python to generate SOW documents using AWS Bedrock.
- ➢ Leveraged AWS Bedrock to automate and streamline the creation of SOWs.
- ➢ Utilized Amazon S3 for scalable data storage and AWS Lambda for serverless processing.
- ➢ Integrated API Gateway for secure and efficient API management.
- ➢ Showcased the potential to enhance productivity and accuracy in creating business documents.

4. **PoC Development for Cheque Fraud Detection on Serverless Cloudera on AWS**

- ➢ Developed an end-to-end PoC for cheque fraud detection as part of the Cloudera for AWS Box Program, demonstrating a serverless Cloudera architecture on AWS.
- ➢ Built an automated pipeline using Python and PySpark to process cheque images, performing AI-based text extraction followed by rule-based fraud detection using Cloudera Data Engineering.
- ➢ Designed data storage using Cloudera Operational Database for cheque metadata and Cloudera Data Warehouse with partitioned Hive tables for fraud result analytics.
- ➢ Enabled real-time fraud monitoring dashboards using Impala and Cloudera Data Visualization, providing continuous visibility into suspicious cheque activity.
- ➢ Showcased the solution to Cloudera stakeholders, highlighting scalability, real-time insights, and suitability for cloud-native fraud detection workflows.

Personal Details:

Name : Bhavesh Chaudhari
Do you have a Passport (Yes/No) : Yes
Visa Status : No